# MLE via Newton–Raphson with Bootstrap Confidence Intervals

MTH210 – Statistical Computing | Project Report


**Name:** Isha Goel
**Roll No.:** 251080068
**File No.:** 21


## 1. Statistical Model

The observed sample $x_1, \ldots, x_n$ ($n = 25$, **File No. 21**) arises from the *Exponentiated Exponential* distribution with shape $\alpha > 0$ and rate $\lambda > 0$. The density and distribution function are

$$f(x; \alpha, \lambda) = \alpha\lambda\, e^{-\lambda x}\left(1 - e^{-\lambda x}\right)^{\alpha-1}, \qquad x > 0, \tag{1}$$

$$F(x; \alpha, \lambda) = \left(1 - e^{-\lambda x}\right)^{\alpha}, \qquad x > 0. \tag{2}$$

Setting $\alpha = 1$ , this reduces to the exponential law.

## 2. Maximum Likelihood Estimation

### 2.1. Log-likelihood and profile reduction

Given an i.i.d. sample, the log-likelihood function is

$$\ell(\alpha, \lambda) = n \ln \alpha + n \ln \lambda - \lambda \sum_{i=1}^{n} x_i + (\alpha - 1) \sum_{i=1}^{n} \ln\left(1 - e^{-\lambda x_i}\right). \tag{3}$$

The partial derivative with respect to $\alpha$ yields a clean closed form:

$$\frac{\partial \ell}{\partial \alpha} = \frac{n}{\alpha} + \sum_{i=1}^{n} \ln\left(1 - e^{-\lambda x_i}\right) = 0, \tag{4}$$

giving the estimator as

$$\hat{\alpha}(\lambda) = \frac{-n}{\displaystyle\sum_{i=1}^{n} \ln\left(1 - e^{-\lambda x_i}\right)} \tag{5}$$

which is strictly positive since each log term is negative for $x_i > 0$. Substituting (5) into (3) converts the two-parameter optimisation into a one-dimensional root-finding problem: find $\lambda$ satisfying

$$g(\lambda) := \frac{n}{\lambda} - \sum_{i=1}^{n} x_i + \left(\hat{\alpha}(\lambda) - 1\right) \sum_{i=1}^{n} \frac{x_i\, e^{-\lambda x_i}}{1 - e^{-\lambda x_i}} = 0. \tag{6}$$

### 2.2. Second-order structure and uniqueness of the MLE

The Hessian matrix $H = \nabla^2 \ell(\alpha, \lambda)$ has entries

$$H_{11} = -\frac{n}{\alpha^2}, \qquad H_{12} = H_{21} = \sum_{i=1}^{n} \frac{x_i\, e^{-\lambda x_i}}{1 - e^{-\lambda x_i}}, \tag{7}$$

$$H_{22} = -\frac{n}{\lambda^2} - (\alpha - 1) \sum_{i=1}^{n} \frac{x_i^2\, e^{-\lambda x_i}}{\left(1 - e^{-\lambda x_i}\right)^2}. \tag{8}$$

Since $\alpha$ is eliminated via profiling, the Newton–Raphson update uses only $g'(\lambda) = H_{22}$. Global uniqueness of the MLE follows from the negative definiteness of $H$,

- $H_{11} = -n/\alpha^2 < 0$ for all $n \geq 1$ and $\alpha > 0$.
- $H_{22} < 0$, and the profile log-likelihood is unimodal in $\lambda$, so $\det(H) = H_{11}H_{22} - H_{12}^2 > 0$ at the unique stationary point.

Together these conditions confirm $H \prec 0$ at $(\hat{\alpha}, \hat{\lambda})$, so the log-likelihood is strictly concave there and the MLE is the unique global maximum.

## 3. Newton–Raphson Implementation

### Update rule

Each iteration applies a Newton step to $g$, then immediately re-evaluates $\alpha$ through the profile formula:

$$\lambda^{(t+1)} = \lambda^{(t)} - \frac{g\left(\lambda^{(t)}\right)}{g'\left(\lambda^{(t)}\right)}, \qquad \alpha^{(t+1)} = \hat{\alpha}\left(\lambda^{(t+1)}\right).$$

Convergence near $\hat{\lambda}$ is quadratic, so the algorithm is efficient once a reasonable starting point is supplied.

### Starting values, termination, and convergence

**Initial guess.** When $\alpha = 1$ the median of the given distribution equals $\ln 2/\lambda$, suggesting

$$\lambda^{(0)} = \frac{\ln 2}{\operatorname{median}(\mathbf{x})}.$$

For the File 21 data set, $\operatorname{median}(\mathbf{x}) \approx 1.225$, giving $\lambda^{(0)} \approx 0.5658$ and $\alpha^{(0)} = \hat{\alpha}(\lambda^{(0)}) \approx 1.3276$.

**Stopping rule.** Iterations continue until the total movement in parameter space drops below $\varepsilon = 0.001$:

$$\Delta^{(t)} = \left|\lambda^{(t+1)} - \lambda^{(t)}\right| + \left|\alpha^{(t+1)} - \alpha^{(t)}\right| < \varepsilon = 0.001$$

with a hard cap of $t_{\max} = 100$ iterations.

**The algorithm converged in 22 iterations.**

**Algorithm 1** Newton–Raphson for the Distribution

**Require:** $\mathbf{x} = (x_1, \ldots, x_n),\ \ \varepsilon = 0.001,\ \ t_{\max} = 100$
**Ensure:** $\hat{\lambda}$, $\hat{\alpha}$, iteration count $t$
$\lambda \leftarrow \ln 2 \,/\, \mathrm{median}(\mathbf{x});\quad \alpha \leftarrow -n \,/\, \sum_i \ln(1 - e^{-\lambda x_i})$
$t \leftarrow 0;\quad \Delta \leftarrow \infty$
**while** $\Delta > \varepsilon$ **and** $t < t_{\max}$ **do**
    Compute $g$ using (6) and $g' = H_{22}$ using (8)
    $\lambda' \leftarrow \lambda - g/g';\quad \alpha' \leftarrow -n \,/\, \sum_i \ln(1 - e^{-\lambda' x_i})$
    $\Delta \leftarrow |\lambda' - \lambda| + |\alpha' - \alpha|$
    $(\lambda, \alpha, t) \leftarrow (\lambda', \alpha', t + 1)$
**end while**
**if** $t = t_{\max}$ **then warn** "Maximum iterations reached without convergence"
**end if**
**return** $\hat{\lambda} \leftarrow \lambda,\ \ \hat{\alpha} \leftarrow \alpha,\ \ t$

### MLE results

| Parameter | Symbol | Start | MLE | Iterations |
|---|---|---|---|---|
| Rate | $\lambda$ | 0.5658 | 1.5549 | **22** |
| Shape | $\alpha$ | 1.3276 | 4.3047 | |

Table 1: Newton–Raphson results for File No. 21 ($n = 25$).

## 4. Bootstrap Confidence Intervals

### 4.1. Two resampling strategies

Let $\hat{\theta} = (\hat{\lambda}, \hat{\alpha})$ be the MLE. For $B = 1000$ **bootstrap replicates** (seed fixed at `set.seed(123)`), each replicate re-estimates $(\lambda, \alpha)$ via the same NR procedure. The 95% percentile interval is

$$\widehat{\mathrm{CI}}_{95\%} = \Big(Q^*_{0.025},\ Q^*_{0.975}\Big),$$

where $Q^*_p$ is the $p$-th empirical quantile of the $B$ replicate estimates.

**NON-PARAMETRIC SAMPLING**

Each bootstrap sample $\mathbf{x}^*$ is drawn *with replacement* from the original $n$ observations. This approach is distribution-free and treats the empirical distribution as the population.

**PARAMETRIC SAMPLING**

Each bootstrap sample is generated from the fitted model $F(\cdot; \hat{\alpha}, \hat{\lambda})$ using the inverse-CDF method: if $U \sim \mathrm{Uniform}(0, 1)$, then

$$X^* = -\frac{1}{\hat{\lambda}} \ln\big(1 - U^{1/\hat{\alpha}}\big)$$

is an exact draw from the fitted distribution.

### 4.2. Results

| **Bootstrap type** | **Parameter** | **2.5% quantile** | **97.5% quantile** |
|---|---|---|---|
| Non-parametric | $\lambda$ | 1.2124 | 2.3526 |
| Non-parametric | $\alpha$ | 2.7406 | 11.9441 |
| Parametric | $\lambda$ | 1.1143 | 2.5148 |
| Parametric | $\alpha$ | 2.4072 | 12.1161 |

Table 2: 95% bootstrap CIs for both parameters ($B = 1000$, `set.seed(123)`).

Both methods yield broadly consistent intervals, reinforcing the reliability of the MLE. The parametric CIs are somewhat wider: by drawing from the *estimated* model rather than the data directly, they absorb the additional uncertainty introduced by plugging in $(\hat{\alpha}, \hat{\lambda})$ in place of the true parameters. The non-parametric CIs are tighter since resampling is restricted to the observed empirical support. The wide range for $\alpha$ under both methods is expected for $n = 25$, as the shape parameter is sensitive to individual observations.

## 5. Summary

Newton–Raphson was applied to the score equation with a median-based starting point. The combined absolute-change criterion ($\varepsilon = 0.001$) was met after **22 iterations**, yielding

$$\hat{\lambda} = 1.5549, \qquad \hat{\alpha} = 4.3047.$$

Negative-definiteness of the Hessian guarantees these estimates are unique global maximisers of the log-likelihood. Bootstrap intervals at the 95% level from both resampling strategies are mutually consistent; the wider parametric intervals reflect the extra model uncertainty inherent in that approach.

**Summary of Key Results (File No. 21)**

**File No.:** **21**

**Initial guess:** $\lambda^{(0)} \approx \mathbf{0.5658}$, $\alpha^{(0)} \approx \mathbf{1.3276}$

**Stopping criterion:** $\Delta^{(t)} < \mathbf{0.001}$, $t_{\max} = \mathbf{100}$

**Iterations to converge:** **22**

**MLE:** $\hat{\lambda} = \mathbf{1.5549}$, $\hat{\alpha} = \mathbf{4.3047}$

**Bootstrap replicates:** $B = 1000$ (`set.seed(123)`)

| **Method** | **Param.** | **2.5%** | **97.5%** |
|---|---|---|---|
| Non-parametric | $\lambda$ | 1.2124 | 2.3526 |
| Non-parametric | $\alpha$ | 2.7406 | 11.9441 |
| Parametric | $\lambda$ | 1.1143 | 2.5148 |
| Parametric | $\alpha$ | 2.4072 | 12.1161 |